

RAG, Multimodal AI & Document Intelligence

Research objective

- Understand how the workbench can transform messy enterprise documents and visual information into usable evidence for agentic workflows.

Research areas

Enterprise retrieval

- Local RAG
- Enterprise RAG
- Offline RAG
- Air-gapped RAG
- Semantic retrieval
- Keyword retrieval
- Hybrid retrieval
- Embeddings
- Reranking
- Metadata filtering
- Hierarchical retrieval
- Multi-document retrieval
- Source attribution
- Grounded generation

Document intelligence

- PDF parsing
- Scanned PDF processing
- OCR
- Document layout analysis
- Table extraction
- Document structure extraction
- Technical document understanding
- Document-to-structured-data extraction

Multimodal processing

- Vision-language models
- Scanned documents
- Handwriting
- Photographs

- Engineering drawings
- P&IDs
- Diagrams
- Visual question answering
- Diagram understanding
- Multimodal retrieval
- Multimodal RAG

Failure research

Investigate:

- OCR errors
- Missing information
- Incorrect layout interpretation
- Table extraction errors
- Diagram interpretation errors
- VLM hallucination
- Retrieval failures
- Incorrect chunking
- Wrong document retrieval
- Missing citations/evidence
- Stale knowledge
- Contradictory documents

Keywords

R4 — Technical Research

Local Knowledge Retrieval

- local enterprise RAG architecture
- offline RAG architecture
- air-gapped RAG
- private RAG
- enterprise knowledge retrieval
- local semantic search
- enterprise hybrid search
- metadata-filtered retrieval
- document reranking
- local embedding models
- local reranker models
- retrieval pipeline architecture
- contextual retrieval

- hierarchical document retrieval
- multi-document reasoning
- source-grounded enterprise AI

Document Intelligence

- local document intelligence
- scanned PDF understanding
- technical document understanding
- engineering document understanding
- document layout analysis
- table extraction from technical documents
- document structure extraction
- PDF OCR pipeline
- OCR plus LLM pipeline
- OCR plus VLM pipeline
- document-to-structured-data extraction

Multimodal AI

- local vision-language models
- multimodal document understanding
- engineering drawing understanding
- technical drawing AI
- P&ID understanding AI
- diagram understanding vision-language model
- image-to-structured-information extraction
- handwritten document understanding
- industrial photograph understanding
- scanned engineering document analysis
- multimodal RAG
- vision-language RAG
- local OCR vision pipeline

Artifact Generation

- local AI document generation
- AI-generated Word documents
- AI-generated engineering reports
- AI-generated approval notes
- AI-generated PowerPoint presentations
- AI-generated Excel spreadsheets
- structured artifact generation
- programmatic business document generation
- AI artifact validation

- artifact consistency verification

Required conclusion

The research must establish:

Raw enterprise information



Processing



Structured representation



Retrieval



Evidence



Reasoning



Agent action

and identify the failure points at every stage.